



(no subject)  
3 messages

vinod@keralauniversity.ac.in <vinod@keralauniversity.ac.in>  
To: vinodchandrass@gmail.com

Wed, Sep 30, 2026 at 5:04 PM

Yes. Based on your research profile—**nature-inspired optimization (SDA, Phototropic Algorithm)**, evolutionary/multi-objective optimization, AI/ML, SDN, edge intelligence, and engineering applications—I see several possibilities for an IEEE Transactions-level review.

I checked recent IEEE Transactions review papers first. Importantly, **IEEE Transactions on Evolutionary Computation (TEVC)** is publishing substantial survey work, including recent surveys on MetaBBO and evolutionary computation for general-purpose AI.

My strongest suggestion

“From Metaheuristics to Autonomous Optimizers: A Systematic Review of Learning-Driven Algorithm Design”

Target: *IEEE Transactions on Evolutionary Computation*

This would be more ambitious and current than another conventional “survey of metaheuristic algorithms.”

The central question would be:

How is AI changing the way optimization algorithms themselves are designed, adapted, selected, and controlled?

The review could connect:

Classical metaheuristics → adaptive metaheuristics → hyper-heuristics → automated algorithm configuration → reinforcement-learning-based optimization → MetaBBO → LLM-assisted optimizer design

This is particularly timely because IEEE TEVC has just published a 2026 survey on **Meta-Black-Box Optimization**, covering algorithm selection, configuration, solution manipulation and algorithm generation, including reinforcement learning and LLMs.

So I would **not** simply duplicate that paper. Instead, your article should investigate the broader transition from **hand-designed optimization algorithms to self-adaptive/autonomous optimization systems**, with a stronger engineering perspective.

Three topics I would seriously consider

| Topic                                                                                           | Possible IEEE journal | Novelty potential | Fit with your work |
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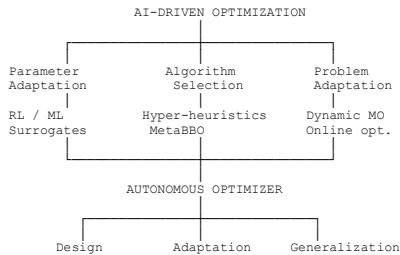
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A possible title:

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This would allow you to bring together your experience with **SDA, Phototropic Algorithm, evolutionary algorithms, multi-objective optimization and hyper-heuristics** without making the paper merely a review of your own algorithms.

A useful taxonomy could be:



The article could then introduce a “**degree of autonomy**” framework, for example:

- Level 0 — fixed algorithm
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That conceptual framework could be the **original contribution of the review**, rather than merely collecting papers.

2. Learning-driven multi-objective optimization

This is probably the **closest to your current research**.

Possible title:

Learning-Driven Multi-Objective Evolutionary Optimization: A Systematic Review of Adaptation, Surrogates, and Knowledge Transfer

Sections could cover:

- Adaptive MOEAs
- Preference learning

- Surrogate-assisted optimization
- Reinforcement learning
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Your **KOA/MOEA work** would give you useful expertise for the experimental methodology, benchmarking and statistical-analysis sections.

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This connects particularly well with your **IoT-based anomalous driving behaviour detection** work.

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Instead, the novelty could be the intersection:

**Edge AI + driver behaviour + real-time anomaly detection + explainability + privacy + resource constraints + safety**

Your ESP32/MPU6050/XGBoost/SHAP work could provide an **illustrative reference architecture**, while the review remains broad.

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**“Optimization for Edge Intelligence: A Taxonomy of Learning, Resource Allocation, Routing and Autonomous Decision-Making”**

Potential target:

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The scope could integrate:

- Edge-cloud optimization
- AI model placement
- Task offloading
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- SDN optimization
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and systematically analyse approximately **300–500 papers from 2015–2026**, with:

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This is especially relevant because TEVC is currently publishing work at precisely the intersection of **evolutionary computation, automated algorithm design and AI**.

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I think this is **more promising for your profile than another review specifically on SDA, phototropic algorithms, or generic nature-inspired algorithms**, because it uses your optimization background while positioning the paper around a much broader emerging research question.

Yes. I have now done the **final audit**, and I would change my recommendation again rather than force the earlier “AI-native computing” idea.

### The important finding

The broad direction we were considering is **already too well covered**:

- ACM CSUR has a comprehensive 2025 survey of on-device/edge AI.
- Edge-AI architecture/software/hardware has several 2026 surveys.
- Heterogeneous computing and DPU/SmartNICs have a 2026 survey.
- Physical AI now has a TPAMI survey, plus several 2026 surveys.
- World models for Physical AI have a dedicated September 2026 survey.
- Digital twins + Physical AI have already been reviewed.
- Even **continual learning in Edge AI** has now appeared in IEEE TNSE in 2026.
- Lifelong learning for LLM agents has already reached TPAMI.

So I would not start an article titled anything like “AI-Native Computing”, “Physical AI”, “Edge AI”, or “Continual AI”.

### What I would start instead

I think we should move **one level above the individual technologies**.

**My current strongest candidate:**

## Adaptive AI Systems in the Open World: A Systematic Review of Lifelong Adaptation, Resource Adaptation, and Safety

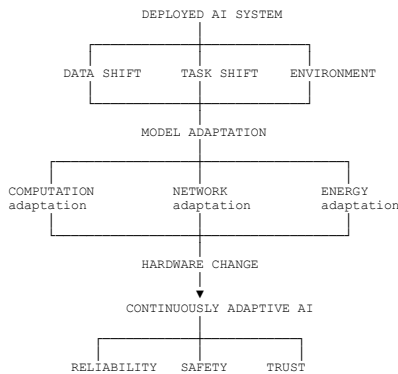
This is deliberately different from conventional continual learning.

The central question is:

**How can an AI system remain reliable when the environment, data, tasks, resources, hardware, users, and objectives all change after deployment?**

That is a **10-year problem**.

And it connects several research streams that are usually reviewed separately:



### Why I find this more interesting

Existing surveys tend to isolate the problem:

- **Continual learning** → model/data adaptation
- **Edge AI** → resource-constrained deployment
- **Federated learning** → distributed learning
- **Lifelong agents** → agent adaptation
- **Physical AI** → physical interaction
- **AI orchestration** → infrastructure adaptation

For example, the 2026 IEEE TNSE review of continual learning in Edge AI explicitly identifies **communication-learning co-design** and sustainable long-term operation as gaps.

But that still treats adaptation primarily as **continual learning**.

**The larger unanswered question is:**

**What does “adaptation” mean for an AI system as a whole?**

Not merely:

How does the model learn new data?

But:

How does the *entire deployed intelligence system* adapt?

That includes:

**model + data + memory + compute + network + energy + sensors + task + environment + safety constraints.**

That is the gap I would investigate next.

## The potential original framework

I would propose a framework called something like:

### 6-Dimensional Adaptive AI

Dimension

What changes?

|             |                                        |
|-------------|----------------------------------------|
| Data        | Distribution, quality, availability    |
| Task        | Objective, workload, user requirements |
| Model       | Parameters, architecture, knowledge    |
| Resources   | CPU/GPU/NPU, memory, energy            |
| Environment | Physical/network/context conditions    |
| Constraints | Safety, privacy, latency, reliability  |

Then ask:

**Can existing AI systems adapt simultaneously across these dimensions?**

That is much more interesting than another survey of continual-learning algorithms.

## Why this could become a high-impact review

The paper could establish a new concept:

### System-level AI adaptation

Most current work optimizes:

#### model adaptation

but next-generation deployed AI will require:

#### system adaptation.

For example, suppose an autonomous vehicle’s environment changes.

It may need to simultaneously:

- adapt its model to weather changes;
- change sensor fusion;
- alter inference frequency;
- move computation between vehicle and edge;
- change communication strategy;
- change energy allocation;
- modify its planning policy;
- detect whether adaptation itself is safe.

That is **not simply continual learning**.

## And this fits your background surprisingly well

You have work spanning:

- machine intelligence
- optimization
- SDN/network optimization
- edge/IoT AI
- real-time driver behaviour detection
- adaptive algorithms
- multi-objective optimization

So you could understand the different layers without making the paper about your own previous algorithms.

Your **optimization background** could become particularly valuable because adaptation is fundamentally a **multi-objective dynamic optimization problem**:

accuracy × latency × energy × communication × cost × safety × privacy.

That gives you a distinctive perspective compared with a conventional ML survey.

## But I would NOT start writing yet

I want to be stricter this time.

Before committing, I would perform one final search specifically for:

**“system-level adaptive AI / adaptive AI systems / lifelong adaptive systems / self-adaptive AI / open-world AI / dynamic AI systems / deployment-time adaptation”**

across **Scopus, Web of Science, IEEE Xplore, ACM DL, ScienceDirect and Springer**, and classify every relevant review into:

**Data adaptation | Model adaptation | Agent adaptation | Hardware adaptation | Network adaptation | Resource adaptation | Safety adaptation | System-level adaptation**

If we discover that somebody has already done essentially the same synthesis, **we abandon this topic too**.

If the matrix shows that the literature is fragmented, then we have something much stronger:

### Proposed final paper concept

**Adaptive AI in the Open World: A Systematic Review and Taxonomy of Lifelong, Resource-Aware, and Safe AI System Adaptation**

And the central contribution would be a **system-level adaptation taxonomy**, not another survey of algorithms.

**This is the direction I would investigate next before you invest any writing effort.**

Sent from my iPhone

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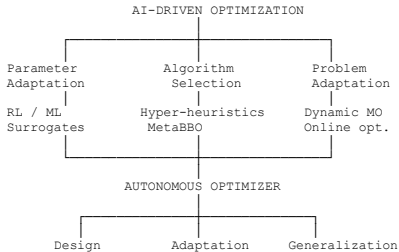
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Potential target:

**IEEE Transactions on Network Science and Engineering**

The scope could integrate:

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and systematically analyse approximately **300–500 papers from 2015–2026**, with:

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**A stronger final title**

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I think this is **more promising for your profile than another review specifically on SDA, phototropic algorithms, or generic nature-inspired algorithms**, because it uses your optimization background while positioning the paper around a much broader emerging research question.

Sent from my iPhone

Yes. **It is absolutely possible to start such an article now**, but I would not start writing the manuscript yet.

From our discussion, my recommendation is to pursue a **forward-looking, high-impact review at the intersection of AI and the physical/digital computing infrastructure**, rather than another generic AI/ML survey.

My preferred direction

AI-Native Computing: From Edge Intelligence to Physical AI, Autonomous Systems, and Heterogeneous Computing

But I would **not use this title yet**. It is a research direction, not the final title.

The reason I like this direction for you is that it connects several technologies that are likely to remain important over the next decade:

Foundation models → edge AI → heterogeneous accelerators → 6G → digital twins → robotics/physical AI → autonomous systems

It also allows you to use your existing strengths in **AI, optimization, edge computing, SDN/networking and machine intelligence**, without making the review narrowly about one of your existing algorithms.

However, there is an important condition

Before committing, I would perform one final “**publication-gap audit**”:

- Scopus
- Web of Science
- IEEE Xplore
- ACM Digital Library
- ScienceDirect
- Springer
- Wiley
- Nature
- 2025–2026 publications

and identify approximately **50–100 closest reviews**.

Then I would construct:

| Question                                                   | What we need to establish |
|------------------------------------------------------------|---------------------------|
| Is the technology important for 2030–2035?                 | Yes                       |
| Is the primary literature growing rapidly?                 | Yes                       |
| Are existing reviews fragmented?                           | Ideally yes               |
| Is there a specific missing synthesis?                     | Essential                 |
| Can we propose a new taxonomy?                             | Essential                 |
| Can we analyze hundreds of papers systematically?          | Yes                       |
| Can we identify research roadmaps?                         | Yes                       |
| Can the paper offer something beyond a literature summary? | Essential                 |

If the answer remains positive, **then start the article**.

My advice

I would aim first at:

ACM Computing Surveys

or

IEEE Transactions on Artificial Intelligence

rather than selecting a journal first and then trying to fit a topic into it.

And I would make the paper a **systematic + bibliometric + conceptual review**, with an original framework and research roadmap.

**So yes—this is a project worth starting. But the next step should be the final gap audit, not drafting Section 1.** Once we identify the precise untouched gap, I can help you build the complete paper around it.

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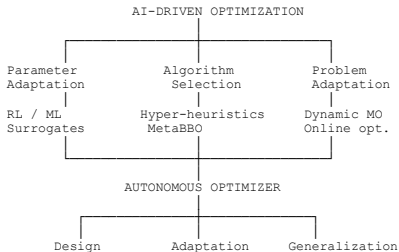
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